

Stock and Flow 101

Guiding Problem Book and Master Reference

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Finite balances first; rates second; limits only when needed.

How to use this guide. Choose the system, identify the stock, identify the net flow (in (+) and out (-)), write the balance, and only then introduce a rate if it is useful.

1 Defining Diagram and Standard Translation

The basic notation is

$$\boxed{S_i \xrightarrow{\downarrow \mathcal{F}} S_f} \quad (1)$$

where S_i and S_f are the initial and final stocks, respectively, and $\mathcal{F}$ is the net inflow accumulated during the transition.

The diagram is translated algebraically as

$$S_i + \mathcal{F} = S_f, \quad \text{or equivalently} \quad \mathcal{F} = S_f - S_i. \quad (2)$$

Thus the net inflow $\mathcal{F}$ is determined by the initial and final stocks. We call $S_f - S_i$ the final-minus-initial difference, or simply the gain in S .

Path independence and state quantities

Because S is a stock—a state quantity—its gain

$$\Delta S := S_f - S_i$$

is determined by the two endpoint states. This does not mean that every transfer occurring during a process is path-independent.

A transfer W is path-independent if its value between two states depends only on the initial and final states, not on the intermediate process. When such a transfer is viewed as an outflow from the system, we may introduce a corresponding stock U using the loss form

$$\boxed{U_i \xrightarrow{\downarrow W} U_f} \quad (3)$$

which is translated algebraically as

$$U_i - W = U_f, \quad W = U_i - U_f = -\Delta U. \quad (4)$$

Thus path independence allows a transfer quantity to be represented by the difference of a corresponding state quantity.

Question 0. Consult a standard physics textbook and find the definition of potential energy in terms of the work done by a conservative force.

Translate the textbook statement into the outward stock–flow diagram in (3). What are the stock and the outflow? What does the diagram say algebraically? Find a non-conservative example

Answer. In (3), stock: U and outflow: W). Thus, by (4), $W = -\Delta U$. Frictional work gives a non-conservative counterexample.

Time labels and average rate

If the two states are also labeled by times $t_i < t_f$, with $S_i = S(t_i)$ and $S_f = S(t_f)$, set $\Delta t = t_f - t_i > 0$. The average net flow rate over the interval $[t_i, t_f]$ is defined by

$$\bar{I}_{[t_i, t_f]} := \frac{\Delta S}{\Delta t} = \frac{S_f - S_i}{t_f - t_i}.$$

Hence, (1) may be written as

$$\boxed{S(t_i) \xrightarrow{\downarrow \bar{I} \Delta t} S(t_f)}.$$

No assumption has been made that the instantaneous flow rate is constant. The average rate $\bar{I}$ is defined exactly by the finite gain ΔS and the elapsed time Δt .

Three levels

The bookkeeping has three distinct levels:

$$\begin{aligned} \text{stock} &: S, \\ \text{finite flow or gain} &: \mathcal{F} = S_f - S_i, \\ \text{average flow rate} &: \bar{I} = \frac{\mathcal{F}}{\Delta t}. \end{aligned}$$

Their units satisfy

$$[\mathcal{F}] = [S], \quad [\bar{I}] = [S]/[t].$$

★ Calculus bridge: the continuous-time reading

The finite balance is exact and requires no limiting process. Calculus supplies a local description of the same bookkeeping.

If S is defined near t and the following limit exists, define the instantaneous net flow rate at t by

$$I(t) = \lim_{h \rightarrow 0} \frac{S(t+h) - S(t)}{h} = \frac{dS}{dt}(t). \quad (5)$$

For readers familiar with dynamical systems: assigning continuous-time flow-rate functions to a stock–flow diagram gives an ordinary differential equation. The diagram is therefore not merely an illustration of the equation; it is a bookkeeping syntax from which the dynamical law can be read.

Finance/accounting examples

Question 1. A checking account begins with \$1,200. During the next 14 days, \$1,800 enters and \$1,200 leaves. Find the final balance, the gain, and the average net cash-flow rate.

Answer. The net transition is represented by

$$B_i \xrightarrow[\downarrow \$1200]{\downarrow \$1800} B_f$$

so the net flow is $\mathcal{F} = \$1800 - \$1200 = \$600$. Hence

$$B_f = B_i + \mathcal{F} = \$1200 + \$600 = \$1800,$$

and

$$\bar{I} = \frac{\$600}{14 \text{ days}} \approx \$42.86/\text{day}.$$

This does not mean that \$42.86 entered every day.

Question 2. Treat outstanding debt D as a stock. During one billing period there is new borrowing, interest, fees, and a payment. Write the finite stock–flow balance.

Answer. The finite bookkeeping is exact before any compounding model is introduced:

$$D_i \xrightarrow[\downarrow \text{payments}]{\downarrow \text{new borrowing, interest, and fees}} D_f$$

Thus

$$D_f - D_i = (\text{new borrowing}) + (\text{interest/fees}) - (\text{payments}).$$

Exact finite logic

In the basic model with no internal source or sink,

$$\mathcal{F} = 0 \iff S_f = S_i.$$

Thus zero net finite flow means that the stock has no net gain over the chosen interval, namely $\Delta S = 0$. It does not imply that no gross inflow or outflow occurred during the interval.

Question 3. A savings account begins at \$4,000. During a 30-day month, \$1,500 is deposited, \$900 is withdrawn, and \$20 is credited as interest. Find the final balance, gain, and average net flow rate.

Answer.

$$B_i \xrightarrow[\downarrow \$900]{\downarrow \$1500 \quad \downarrow \$20} B_f$$

Therefore

$$\mathcal{F} = \$1500 + \$20 - \$900 = \$620.$$

$$B_f = B_i + \mathcal{F} = \$4000 + \$620 = \$4620,$$

and

$$\bar{I} = \frac{\$620}{30 \text{ days}} \approx \$20.67/\text{day}.$$

Again, the average rate summarizes the endpoint gain; it need not describe any particular day.

2 General Framework Questions

A1. Diagram. Draw a stock–flow diagram and translate it into an equation.

A2. Gain. Starting from

$$S_i + \mathcal{F} = S_f,$$

solve for $\mathcal{F}$ and identify the gain in S .

A3. Average rate. If time labels are attached to the two states, define

$$\bar{I} = \frac{\mathcal{F}}{\Delta t}.$$

A4. Path independence. What does it mean for a transfer W to be path-independent? Show how a path-independent loss can be represented by a stock U through

$$W = U_i - U_f = -\Delta U.$$

Why can the same construction not automatically be made for a path-dependent transfer? Find a counterexample.

A5. Zero net flow. Show that

$$\mathcal{F} = 0 \iff S_f = S_i.$$

Interpret the result in the basic model with no internal source or sink.

A6. Sign. Explain the meanings of

$$\mathcal{F} > 0 \quad (\text{gain}), \quad \mathcal{F} < 0 \quad (\text{loss}), \quad \mathcal{F} = 0 \quad (\text{balanced}).$$

A7. Stock, flow, and rate. Explain the conceptual and dimensional differences among

$$S, \quad \mathcal{F} = S_f - S_i, \quad \bar{I} = \frac{\mathcal{F}}{\Delta t}.$$

A8. Mechanics realization. Give a Newtonian example and identify stock, finite flow, and average rate.

A9. Finance realization. Give a financial example and identify initial stock, final stock, finite flow, and average flow rate.

Question 4. A tank contains 120 L of water. During an hour, 50 L enters and 30 L leaves. Treat the amount of water as the stock. Find the final stock, gain, and average net flow rate.

Answer.

$$120 \text{ L} \xrightarrow[\downarrow 30 \text{ L}]{\downarrow 50 \text{ L}} S_f$$

Thus

$$S_f = 120 \text{ L} + 20 \text{ L} = 140 \text{ L}, \quad \text{and} \quad \bar{I} = 20 \text{ L h}^{-1}.$$

Physical realizations at a glance

Context	Stock	Flow	Average rate
Linear motion	Momentum $\mathbf{p}$	Impulse $\mathbf{J}$	force $\bar{\mathbf{F}}$
Rotational motion	Angular momentum $\mathbf{L}$	angular impulse	torque $\bar{\boldsymbol{\tau}}$
Energy	energy E, U, K	work/heat W, Q	power $\bar{P}$
Electricity	charge q	charge transfer	current $\bar{I}$
Cash	balance B	net cash flow	cash-flow rate

3 Finite Balance Algebra

A system may have several boundary inflows and outflows. Their signed sum is the net boundary transfer

$$\mathcal{F}_\partial := \sum \mathcal{F}_{\text{in}} - \sum \mathcal{F}_{\text{out}}, \quad S_i \xrightarrow[\downarrow \mathcal{F}_\partial]{\downarrow \mathcal{F}_\partial} S_f \quad (6)$$

where the subscript ∂ reminds us that these transfers cross the boundary of the chosen system.

Internal source and sink

Some stocks may also be produced or destroyed inside the system. Let $\mathcal{G}$ denote the net internal production during the transition, with negative $\mathcal{G}$ representing net internal destruction.

The general finite balance is

$$S_f - S_i = \mathcal{F}_\partial + \mathcal{G}. \quad (7)$$

This equation separates three logically different statements:

$$\begin{aligned} \text{zero accumulation} &: S_f - S_i = 0, \\ \text{isolated system} &: \mathcal{F}_\partial = 0, \\ \text{no internal production/destruction} &: \mathcal{G} = 0. \end{aligned}$$

For a conserved quantity in an isolated system, $\mathcal{F}_\partial = 0$ and $\mathcal{G} = 0$, and therefore

$$S_f = S_i.$$

Telescoping

For intermediate states $S_0, S_1, \dots, S_n$, we have

$$(S_1 - S_0) + (S_2 - S_1) + \dots + (S_n - S_{n-1}) = S_n - S_0. \quad (8)$$

All intermediate stocks cancel. Finite balances therefore compose exactly, without any limiting process.

- B1.** Explain why a finite sequence of deposits and withdrawals can be replaced by one net cash transfer.
- B2.** Explain why zero final gain does not imply zero gross flow.
- B3.** Distinguish “the stock is unchanged” from “the quantity is conserved.”

★ **General continuous-time balance**

If instantaneous rates exist, let I_{∂} denote net boundary inflow rate and G denote net internal production rate. The continuous-time form of (7) is

$$\frac{dS}{dt} = I_{\partial} + G. \quad (9)$$

C1. Three levels. Classify

$$S_f - S_i = \mathcal{F}, \quad S_f - S_i = \bar{I} \Delta t, \quad \frac{dS}{dt} = I.$$

Which one requires a limit?

C2. Units. If $[S] = Q$, show that both $\bar{I}$ and $\frac{dS}{dt}$ have units Q/time .

Reconstructing the finite balance

If S is sufficiently regular and the instantaneous rates are integrable, then

$$S_f - S_i = \int_{t_i}^{t_f} \frac{dS}{dt}(t) dt = \int_{t_i}^{t_f} (I_{\partial}(t) + G(t)) dt. \quad (10)$$

Thus integration reconstructs the finite balance (7) from the rate history.

★ **From a control volume to a local law**

The same bookkeeping can be localized in space. Let ρ be a stock density in a fixed region V , $\mathbf{j}$ be its outward flux density, and σ be its internal production density.

The stock contained in V , net boundary inflow rate, and internal production rate are

$$S_V(t) = \int_V \rho(\mathbf{x}, t) dV, \quad I_{\partial, V} = - \oint_{\partial V} \mathbf{j} \cdot \hat{\mathbf{n}} dA, \quad G_V = \int_V \sigma dV,$$

where $\hat{\mathbf{n}}$ is the outward unit normal to ∂V . The general stock-flow equation (9) becomes

$$\frac{d}{dt} \int_V \rho dV = - \oint_{\partial V} \mathbf{j} \cdot \hat{\mathbf{n}} dA + \int_V \sigma dV. \quad (11)$$

The minus sign appears because $\mathbf{j} \cdot \hat{\mathbf{n}} > 0$ represents outward flow and therefore decreases the stock inside V .

Using the divergence theorem, $\oint_{\partial V} \mathbf{j} \cdot \hat{\mathbf{n}} dA = \int_V \nabla \cdot \mathbf{j} dV$, if the balance (11) holds for every fixed region V , and the fields are sufficiently regular, then

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = \sigma. \quad (12)$$

This is the local stock-flow-production equation.

Local conservation

If the quantity is locally conserved, then $\sigma = 0$, and the local balance (12) reduces to

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0. \quad (13)$$

4 Equation of Motion

Momentum as a Stock

For a chosen object or system, $\mathbf{p} = m\mathbf{v}$, the finite stock-flow statement is

$$\mathbf{p}_i \xrightarrow{\Downarrow \mathbf{J}} \mathbf{p}_f, \quad \mathbf{p}_i \xrightarrow{\Downarrow \bar{\mathbf{F}} \Delta t} \mathbf{p}_f,$$

where the momentum flow $\mathbf{J}$ is called the impulse, and $\bar{\mathbf{F}}$ is the average force:

$$[\mathbf{p}] = [\mathbf{J}] = \text{mkg/sec} = \text{N sec}, \quad [\bar{\mathbf{F}}] = \text{N}.$$

D1. Momentum balance. Write the stock-flow diagram and finite balance for momentum.

D2. Same impulse, different time. Two stopping mechanisms produce the same Δp , but one acts ten times longer. Compare their average-force magnitudes. (Check your seatbelt!)

D3. ★ Instantaneous form. If the local limit exists, explain how the finite balance leads to

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}.$$

Question 5. A 75 kg driver moving at 20 m/sec is brought to rest. Compare the average stopping forces for stopping intervals of 0.30 sec with a seatbelt and 0.015 sec without. Estimate the corresponding masses whose weights equal these force magnitudes, using $|g| = 9.81 \text{ m s}^{-2}$.

Answer.

$$\Delta p = 0 - 75 \text{ kg } 20 \text{ m/sec} = -1500 \text{ mkg/sec}.$$

Hence

$$F_{\text{with}} = \frac{-1500 \text{ kgm/sec}}{0.30 \text{ sec}} = -5.0 \times 10^3 \text{ N}$$

while

$$F_{\text{without}} = \frac{-1500 \text{ kgm/sec}}{0.015 \text{ sec}} = -1.0 \times 10^5 \text{ N}$$

Note that the negative signs show that the average net force on the driver is directed backward. Same impulse, twenty times shorter interval, twenty times larger average-force magnitude.

The corresponding masses are

$$\frac{5.0 \times 10^3 \text{ N}}{9.81 \text{ m s}^{-2}} \approx 510 \text{ kg}, \quad \text{and} \quad \frac{1.0 \times 10^5 \text{ N}}{9.81 \text{ m s}^{-2}} \approx 10\,000 \text{ kg}.$$

Angular Momentum as a Stock

For a chosen system and a chosen origin, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$,

$$\mathbf{L}_i \xrightarrow{\Downarrow \mathbf{J}_\tau} \mathbf{L}_f, \quad \mathbf{L}_i \xrightarrow{\Downarrow \bar{\boldsymbol{\tau}} \Delta t} \mathbf{L}_f,$$

where $\mathbf{J}_\tau$ is the angular impulse, and $\bar{\boldsymbol{\tau}}$ is the average net torque:

$$[\mathbf{L}] = \text{kgm}^2/\text{sec} = \text{Nm sec}, \quad [\bar{\boldsymbol{\tau}}] = \text{N m}.$$

Notice that torque and energy have the same dimensions N m, but they are different physical quantities.

D4. Angular-momentum balance. Draw the stock-flow diagram and identify the stock, finite flow, and average flow rate.

D5. Same angular impulse, different time. Two processes produce the same $\Delta \mathbf{L}$, but one acts five times longer. Compare their average-torque magnitudes.

D6. ★ Instantaneous form. Explain the structural parallel

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}, \quad \boldsymbol{\tau} = \frac{d\mathbf{L}}{dt}.$$

Question 6. A wheel's angular momentum about its axis changes from $2.4 \text{ kgm}^2/\text{sec}$ to $5.4 \text{ kgm}^2/\text{sec}$ during 0.60 sec . Find the angular impulse and the average net torque.

Answer.

$$\Delta L = 5.4 - 2.4 = 3.0 \text{ kgm}^2/\text{sec},$$

so

$$J_{\tau} = \Delta L = 3.0 \text{ Nm sec}, \quad \bar{\tau} = \frac{J_{\tau}}{\Delta t} = 5.0 \text{ N m}.$$

5 Energy as a Stock

Work-Energy Theorem

In the simplest form of a relation between work and energy:

$$K_i \xrightarrow{\Downarrow W_{\text{net}}} K_f, \quad (14)$$

Here the net work is the flow into the kinetic-energy stock. Define average power $\bar{P}$ by

$$K_i \xrightarrow{\Downarrow \bar{P} \Delta t} K_f.$$

Units:

$$[E] = [W] = \text{J}, \quad [P] = \text{J/sec} = \text{W}.$$

E1. Work-energy balance. Identify stock and finite flow; then define average power.

E2. Force and power as rates. Compare

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} \quad \text{and} \quad P = \frac{dE}{dt}.$$

In what structural sense are they analogous?

E3. State, process, and path independence. Explain why K_i and K_f label states while work describes a transfer between states. When does a work quantity define a corresponding potential-energy stock U ?

Question 7. A motor transfers 12 kJ of energy in 4.0 sec . Find the average power. What happens if the same energy transfer takes 12 sec ?

Answer.

$$\frac{12000 \text{ J}}{4.0 \text{ sec}} = 3.0 \text{ kW}, \quad \frac{12000 \text{ J}}{12 \text{ sec}} = 1.0 \text{ kW}.$$

Same energy flow, three times longer interval, one-third the average power.

Potential energy — stored ability to do work

A conservative force has path-independent work:

$$U_i \xrightarrow{\Downarrow W_c} U_f, \quad W_c = -\Delta U.$$

Hence

$$W_c = -\Delta U \implies \bar{P}_c = -\frac{\Delta U}{\Delta t} \implies P_c = -\frac{dU}{dt}$$

Since $W_c = \int_i^f \mathbf{F}_c \cdot d\mathbf{x}$, the mechanical power is $P_c = \mathbf{F}_c \cdot \mathbf{v}$. We therefore obtain

$$dU = -\mathbf{F}_c \cdot d\mathbf{r} \implies \mathbf{F}_c = -\nabla U.$$

In one dimension this becomes

$$F_x = -\frac{dU}{dx}.$$

Standard mechanical potentials

System	Potential energy	Force
Uniform gravity	$U(y) = m g y + C$	$F_y = -m g $
Elastic spring	$U(x) = \frac{1}{2}kx^2 + C$	$F_x = -kx$
Newtonian gravity	$U(r) = -\frac{GMm}{r} + C$	$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}$

Question 8. An ideal spring with spring constant $k > 0$ changes from elongation x_i to x_f . Using the potential-energy stock rather than directly integrating the force, find the work done by the spring.

Answer. For

$$U(x) = \frac{1}{2}kx^2,$$

the stock-flow diagram gives immediately

$$W_{\text{spring}} = U_i - U_f = \frac{1}{2}k(x_i^2 - x_f^2).$$

No information about the intermediate motion is required.

Conservation of Total Mechanical Energy

In (14), write $W_{\text{net}} = W_c + W_{\text{ext}}$, where W_{ext} denotes work done by forces not represented by a potential-energy stock U in (3):

$$U_i \xrightarrow{\Downarrow W_c} U_f$$

$$K_i \xrightarrow{\Downarrow W_c + W_{\text{ext}}} K_f$$

Hence, for the total mechanical energy $E = K + U$,

$$\Delta E = \Delta(K + U) = W_c + W_{\text{ext}} - W_c = W_{\text{ext}}.$$

Thus

$$W_{\text{ext}} = 0 \implies E_i = E_f \quad (\text{conservation of total mechanical energy}).$$

The conservative work is an internal flow of the combined $K + U$ system and therefore cancels.

Mixed Practice and Compact Review

M1. Diagram first. Draw the defining diagram, translate it as $S_i + \mathcal{F} = S_f$, and identify the gain.

M2. One structure, three languages. Write the same finite stock-flow pattern for a cash balance, linear momentum, and kinetic energy.

M3. Conservation diagnostic. In each context, identify the internal and external flows. What must vanish for the chosen total stock to be conserved?

M4. Average does not mean constant. Explain why $S_f - S_i = \bar{I} \Delta t$ is exact even when flow occurs in bursts. Give one physical and one financial example.

M5. Momentum comparison. Two objects experience the same impulse. One receives it in 0.50 sec , the other in 0.050 sec . Compare average-force magnitudes.

M6. Energy comparison. A motor transfers 12 kJ in 4.0 sec . Identify stock, finite flow, average rate, and units. What changes if the same transfer occurs in 12 sec ?